Chapter 16.7: Stokes' Theorem

V3: Generalizations of the FTC. I can state and apply Green's Theorem, Stokes' Theorem and the Divergence Theorem to solve problems in two and three dimensions. I can choose which theorem is appropriate for different integrals. I can compute curl and divergence of vector fields.

Mechanics

1. Consider the three surfaces: S the hemisphere given by $x^2 + y^2 + z^2 = 4$, for $z \geq 0$, P the portion of a paraboloid given by $z = 4 - x^2 - y^2$, for $z \geq 0$, and H the hyperboloid $z^2 = x^2 + y^2 - 4$, for $z \leq 0$. Use Stokes' Theorem to explain why, if $\mathbf{F}$ is a smooth vector field in $\mathbb{R}^3$, we must have
$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, d\sigma = \iint_P (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, d\sigma = -\iint_H (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, d\sigma$$

2. A particle moves along line segments from the origin to the points $(1, 0, 0)$, $(1, 2, 1)$, $(0, 2, 1)$, and back to the origin under the influence of the force field
$$\mathbf{F}(x, y, z) = z^2\mathbf{i} + 2xy\mathbf{j} + 2y^2\mathbf{k}$$
Find the work done by the field on the particle.

3. Compute $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, d\sigma$ where S is the surface $z = e^{-x^2-y^2}$ for $z \geq \frac{1}{e}$, and $\mathbf{F}(x, y, z) = (-y, x, 1)$.

Applications

4. In electrodynamics, *Faraday's Law* relates the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ via the equation $\text{curl}(\mathbf{E}) = \frac{d}{dt}\mathbf{B}$. Integrating both sides with respect to any surface S yields
$$\iint_S \text{curl}(\mathbf{E}) \cdot \mathbf{n}\, d\sigma = -\frac{d}{dt} \iint_S \mathbf{B} \cdot \mathbf{n}\, d\sigma$$
The integral on the right is called the *magnetic flux*. Applying Stokes' theorem to the left gives
$$\int_{\partial S} \mathbf{E} \cdot \mathbf{T}\, ds = -\frac{d}{dt} \iint_S \mathbf{B} \cdot \mathbf{n}\, d\sigma$$
The integral on the left is called the *electric potential*. Interpret the above equation to deduce *Faraday's Law of Induction:*

   One may generate an ________________ by first causing a change in ________________.

   This is the driving principle underpinning an AC current generator. It uses a system of rotating magnets (or related things) to constantly change the ________________, in order to generate ________________.

Extensions

5. Use Stokes' Theorem to evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, d\sigma$, where $\mathbf{F} = \langle x^2z^2, y^2z^2, xyz\rangle$ and S is the part of the paraboloid $z = x^2 + y^2$ that lies inside the cylinder $x^2 + y^2 = 4$, oriented upward.

6. In Section 16.3, you learned that a curl-free vector field $\mathbf{F}$ on a simply-connected domain is necessarily conservative (ie., path-independent). In this exercise, you will justify this statement using Stokes' Theorem

   (a) Fix a curl-free vector field $\mathbf{F}$, and also fix any two points $A, B \in \mathbb{R}^3$ and let C_1 and C_2 be curves connecting A to B. For the moment, assume that C_1 and C_2 do not intersect. Use Stokes' Theorem to show that

   $$\int_{C_1 \cup -C_2} \mathbf{F} \cdot \mathbf{T}\, ds = 0$$

   where $-C_2$ represents the path C_2 traced backwards (ie., from B to A). *[Hint: To apply Stokes' theorem, you need to pick a surface. Draw a picture of the situation to help you find this surface.]*

   (b) You have shown that line integrals along *simple* closed curves vanish. Of course, arbitrary choices of C_2 possibly intersect C_1! Nevertheless, assuming that C_2 can only intersect C_1 finitely many times, show that we still have that

   $$\int_{C_1 \cup -C_2} \mathbf{F} \cdot \mathbf{T}\, ds = 0$$

   [Hint: Draw a picture. Decompose the closed curve $C_1 \cup -C_2$ into a collection of simple closed curves.]

   (c) Conclude that

   $$\int_{C_1} \mathbf{F} \cdot \mathbf{T}\, ds = \int_{C_2} \mathbf{F} \cdot \mathbf{T}\, ds$$

   (d) Where did we use the simply-connected assumption?